Roberto (André) Mossi Milla

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SUMMARY

Software Engineer experienced in AI-powered full-stack platforms, computer vision systems, and real-time simulation environments. Background includes production web applications, artificial intelligence research, cloud integrations, and end-to-end system development across healthcare, industrial automation, and multimodal AI workflows.

WORK EXPERIENCE

Ingeniería Digital, Tegucigalpa, Honduras

Full-Stack Developer / AI Systems Engineer, May 2026 - Present

AI-Assisted Medical Document Generation Platform

- Developed an AI-assisted medical document generation platform for post-surgery reports (IDR), combining OCR extraction pipelines, LLM-powered grammar correction, and automated professional document structuring using Next.js, React, TypeScript, PostgreSQL, AWS S3, Amazon Textract, and OpenAI APIs.
- Architected authenticated document management workflows using Auth0, enabling users to generate, analyze, edit, version, and manage AI-generated reports through a full-stack web platform.
- Designed scalable file-processing pipelines integrating OCR extraction, cloud storage, AWS S3 buckets, and PostgreSQL persistence for intelligent document retrieval and processing.
- Implemented internationalization workflows and automated translation asset generation using CMake to support multilingual deployments across US and Honduran clients.

AI Vehicle Damage Assessment Platform

- Developed an AI-assisted vehicle damage assessment platform utilizing multimodal AI workflows, VIN history analysis, image-based damage detection, and insurance estimation pipelines.
- Integrated computer vision and LLM APIs to analyze uploaded vehicle imagery, validate VIN-related information, compare against historical vehicle records, and support insurance decision workflows.
- Developed data acquisition services using Playwright to collect and process vehicle information required for valuation and insurance analysis workflows.
- Implemented interactive damage annotation and review tools using Konva, allowing users and insurance personnel to identify, visualize, and validate detected vehicle damage regions.
- Developed responsive client-facing and administrative dashboards using React, Next.js, TypeScript, PostgreSQL, AWS S3, and Shadcn/UI.
- Maintained engineering standards and CI/CD workflows using GitHub Actions, Husky, ESLint, Prettier, lint-staged, and Commitizen.

Gannon University, Pennsylvania, United States

Computer Vision Engineer (Project-Based) Industry Partner (NDA), August 2024 - May 2025

- Architected and deployed a real-time industrial traceability system using Ultralytics YOLO to detect and track pallets, automatically generating structured events with images and detection metadata.[4]
- Collaborated in a cross-functional team contributing across hardware, backend, and software components; took primary ownership of the monitoring dashboard and visualization interface.
- Developed a full-stack web dashboard (React, JavaScript) integrated with backend services and TimescaleDB to visualize detection events, images, and operational analytics in real time.
- Implemented administrative tools within the dashboard to delete images, correct detections, and curate labeled datasets used for iterative object detection model retraining, improving dataset quality and system reliability.
- Engineered an event-driven pipeline that synchronized frame capture with detection metadata, enabling downstream anomaly and damage analysis workflows.
- Delivered recurring technical presentations and system demonstrations to industry stakeholders and internal reviewers, supporting continued project funding and informing architectural decisions, including migration to NVIDIA Jetson hardware for improved performance and thermal stability.
- Containerized distributed services using Docker and integrated multi-language components (C++, Python, Rust) to coordinate inference, data processing, and system communication.

Research Assistant, August 2024 - May 2025

- Conducted research under faculty supervision on multi-agent artificial intelligence simulating natural selection and survival behaviors in dynamic environments.[8]
- Implemented NeuroEvolution of Augmenting Topologies (NEAT) to evolve neural network architectures and behavioral strategies across generations.
- Developed a 3D multi-agent simulation environment in Unreal Engine featuring procedural world generation, physics-based interactions, and reinforcement learning pipelines using C++, CUDA, and PyTorch.
- Research paper accepted and published by the American Society for Engineering Education (ASEE) and presented at a Sigma Xi conference at Penn State University.[5] [3]

Dinant, Tegucigalpa, Honduras

Full-Stack Developer, June - July 2023

- Led the end-to-end design and development of two production web platforms, conducting requirements gathering, system planning, database architecture design, client presentations, and final deployment.[2]
- Architected and implemented a contract lifecycle management system for a device sales, rental, and repair company. Designed relational SQL schemas to track device state (sold, rented, under repair, returned), service eligibility, country-based filtering, and client records across multiple regions.
- Developed a multi-role web application (Oracle APEX, JavaScript, HTML, CSS) enabling guest contract signing, digital signature capture, automated PDF generation, and administrative dashboards with advanced filtering by date, country, device state, and contract status.
- Engineered a repair workflow system with state transitions, damage logging, cost tracking, time tracking, and daily activity reports to ensure accountability and operational transparency.
- Built a factory maintenance management platform enabling real-time machine status reporting, staged maintenance workflows, automated notifications, repair logs, and material usage tracking with automatic inventory deduction and procurement insight reporting.
- Collaborated directly with international stakeholders, conducted iterative demos, incorporated production feedback, and deployed finalized systems used by operational teams.

EDUCATION

Gannon University, Pennsylvania, United States

Bachelor of Science in Computer Systems Engineering, May 2025[9]

Harvard University, Online

CS50's Introduction to Computer Science, December 2022[10]

Extreme Networks, Online

Introduction to Future Networks, May 2022[1]

PUBLICATIONS

- Mossi, R. A. — "Tag AI-Sandbox," Proceedings of the ASEE Annual Conference, 2025 (peer-reviewed).[5]

TECHNICAL SKILLS

- Languages: C++, TypeScript, JavaScript, Python, Java, SQL, Bash, PowerShell.
- Frontend: React, Next.js, Vite, Tailwind CSS, Shadcn/UI.
- Backend & Cloud: Supabase, PostgreSQL, AWS S3, Amazon Textract, Docker, Auth0, REST APIs.
- AI & Computer Vision: OpenAI APIs, Claude APIs, YOLO, CUDA, PyTorch, NEAT, Reinforcement Learning, OCR Pipelines, Multimodal AI Workflows.
- Developer Tooling: GitHub Actions, Husky, ESLint, Prettier, Commitizen, CMake, Git.
- Operating Systems: Linux, Windows, macOS.
- Languages: Spanish (Native), English (Fluent), Japanese (Intermediate).

AWARDS AND HONORS

- "School Development Award" award obtained for outstanding development of a videogame using Blueprints in Unreal Engine 4, 2020[7]
- "School Development Award" award obtained for outstanding development of a videogame using C++ and Unreal Engine 4, 2016[6]

REFERENCES

[1] Extreme Academy. *Extreme Networks Certification*. Dec. 1, 2022. URL: https://andremossi.vercel.app/experiences/extreme (visited on 02/15/2026).

[2] Rigoberto Funes. *Dinant Recommendation Letter*. Feb. 11, 2026. URL: https://andremossi.vercel.app/PDF/Recommendations/dinant.pdf (visited on 02/15/2026).

[3] lah585. *Tag AI-Sandbox Presentation*. Sigma Xi. Mar. 11, 2025. URL: https://sites.psu.edu/behrendsigmaxi2025/2025/03/11/tag-ai-sandbox/ (visited on 02/15/2026).

[4] Andre Mossi. *Computer Vision Project*. Feb. 15, 2026. URL: https://andremossi.vercel.app/experiences/traceability (visited on 02/15/2026).

[5] Roberto Andre Mossi and Ramakrishnan Sundaram. "Tag AI-Sandbox Paper". In: *2025 ASEE North Central Section (NCS) Annual Conference*. 10.18260/1-2–54692. Marshall University, Huntington, West Virginia: ASEE Conferences, Mar. 2025. URL: https://peer.asee.org/54692.

[6] Dowal School. *Software Engineer 2016 Award*. May 12, 2016. URL: https://andremossi.vercel.app/experiences/dowal2016 (visited on 02/15/2026).

[7] Dowal School. *Software Engineer 2020 Award*. May 12, 2020. URL: https://andremossi.vercel.app/experiences/dowal2020 (visited on 02/15/2026).

[8] Ram Sundaram. *Tag AI-Sandbox Recommendation Letter*. Feb. 20, 2025. URL: https://andremossi.vercel.app/PDF/Recommendations/ai-sandbox.pdf (visited on 02/15/2026).

[9] Gannon University. *Gannon University BS Diploma*. May 10, 2025. URL: https://andremossi.vercel.app/experiences/gannon (visited on 02/15/2026).

[10] Harvard University. *Harvard CS50 Certification*. Dec. 20, 2022. URL: https://andremossi.vercel.app/experiences/harvard (visited on 02/15/2026).